Node Link Network Optimization Report

Model Type: **node_link**

Metric	Value
Status	Optimal
Objective Value	$1,400.00
Total Revenue	$1,400.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$1,400.00
Accepted Demands	4 / 5
Rejected Demands	1 / 5
Acceptance Rate	80.0%
Nodes	10
Edges	16
Total Initial Capacity	535.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	S1 -> R1 (100.0%)
Number of Highly Utilized Edges (≥90%)	2
Most Capacity Expansion	None
Total Lost Revenue	$500.00
Number of Expanded Edges	0

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge S1 -> R1: 100.0% used
- Edge R1 -> R3: 100.0% used

Accepted Demands & Flow Paths

Demand 0: S1 -> D1

Bandwidth: 25.0 units **Revenue:** $400.0
Revenue/BW: $16.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D1	100.00%	25.00

Demand 1: S1 -> D2

Bandwidth: 30.0 units **Revenue:** $450.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R2 -> R5 -> D2	100.00%	30.00

Demand 2: S2 -> D2

Bandwidth: 20.0 units **Revenue:** $300.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S2 -> R2 -> R4 -> D2	100.00%	20.00

Demand 4: S1 -> D3

Bandwidth: 15.0 units **Revenue:** $250.0
Revenue/BW: $16.67/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R4 -> R5 -> D3	100.00%	15.00

Rejected Demands

Idx	Source	Target	Bandwidth	Revenue ($)
3	S2	D3	35.0	500.0

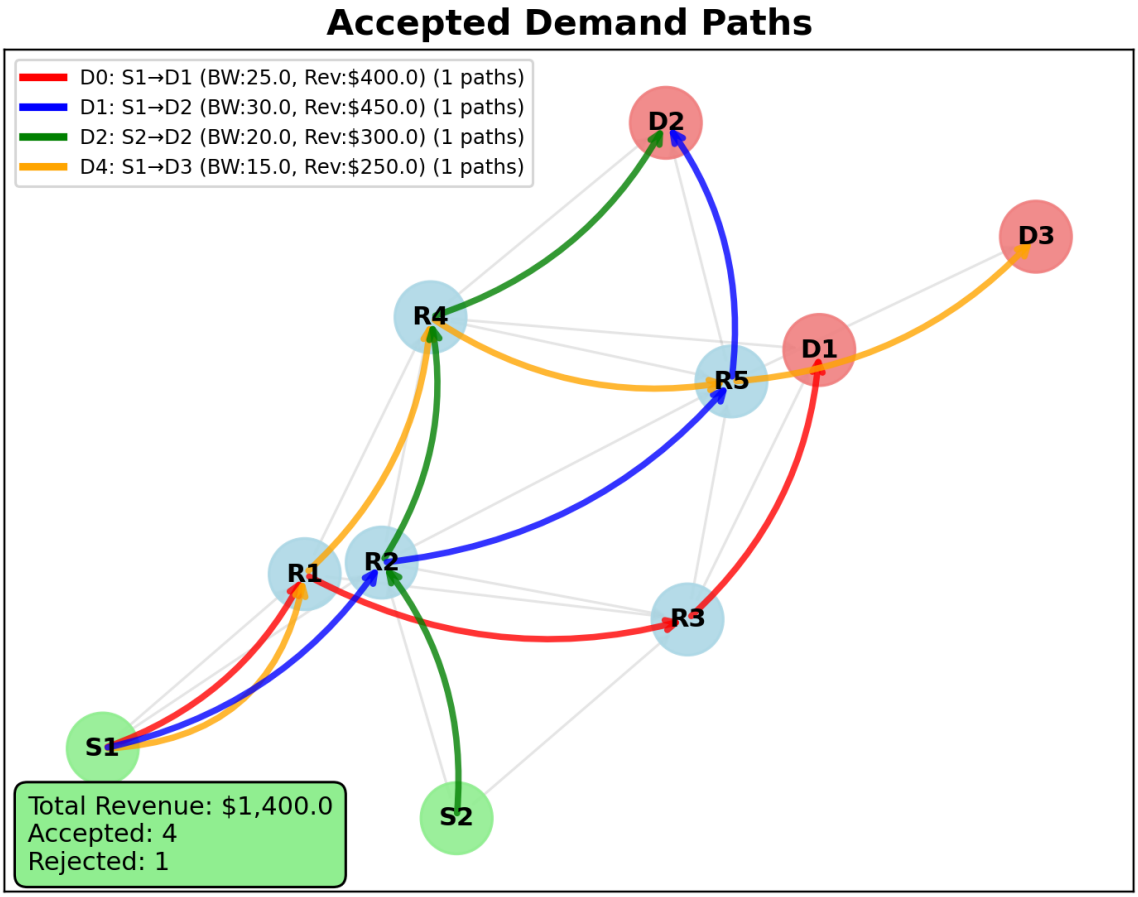
Total Lost Revenue: $500.00

Edge Utilizations (Top 12 Edges)

Edge	Flow	Total Capacity	Utilization (%)
S1 -> R1	40.00	40.00	100.0 %
R1 -> R3	25.00	25.00	100.0 %
S1 -> R2	30.00	35.00	85.7 %
R2 -> R5	30.00	35.00	85.7 %
R2 -> R4	20.00	25.00	80.0 %
R5 -> D2	30.00	40.00	75.0 %
S2 -> R2	20.00	30.00	66.7 %
R4 -> D2	20.00	30.00	66.7 %
R3 -> D1	25.00	40.00	62.5 %
R4 -> R5	15.00	25.00	60.0 %
R1 -> R4	15.00	30.00	50.0 %
R5 -> D3	15.00	50.00	30.0 %

Network Analysis Visualizations

Accepted Demand Paths



Edge Utilization

